

Lecture 3 Exercise Solutions

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1 Important Equations

Probability Density Function

The probability density function (pdf) $f_X(x)$ of a continuous random variable X is a non negative function so $f_X(x) \geq 0$. The pdf is normalized meaning:

$$\int_{-\infty}^{\infty} f_X(x) dx = 1 \quad (1)$$

The probability for the random variable X to be in an interval $[a; b]$ is:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx \quad (2)$$

Cumulative Distribution Function

The cumulative distribution function (cdf) is defined from the probability density function as:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f(x') dx' \quad (3)$$

Mean

The mean of a continuous random variable X is defined as:

$$\mu_X = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad (4)$$

Variance

The variance of a continuous random variable X is defined as:

$$\sigma_X^2 = V[X] = \int_{-\infty}^{\infty} (x - \mu)^2 f_X(x) dx = \int_{-\infty}^{\infty} x^2 f_X(x) dx - \mu_X^2 = E[X^2] - \mu_X^2 \quad (5)$$

From this the standard deviation is $\sigma_X = \sqrt{\sigma_X^2}$. The variance is a non linear operation meaning:

$$V[aX + b] = a^2 \sigma_X^2 + 0 = a^2 \sigma_X^2 \quad (6)$$

Uniform Distribution

For a continuous uniformly distributed random variable in the interval $[a; b]$ $X \sim \text{UD}(a, b)$ the pdf is given as:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

The mean and variance is defined as:

$$\mu_X = E[X] = \frac{a+b}{2}, \quad \sigma_X^2 = V[X] = \frac{(b-a)^2}{12} \quad (8)$$

Normal Distribution

The pdf for a normal distributed continuous random variable $X \sim N(\mu, \sigma^2)$ is defined as:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2 \right] \quad (9)$$

Standard Normal Distribution

The pdf for a standard normal distributed continuous random variable $Z \sim N(0, 1)$ is defined as:

$$f_X(z) = \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} z^2 \right) \quad (10)$$

Where one can standardize a normal distribution as such:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1) \quad (11)$$

Exponential Distribution

An exponential distribution $T \sim \text{Exp}(\lambda)$ describes the distribution of waiting time for the next arrival in a Poisson process:

$$f_T(t) = \lambda e^{-\lambda t}, \quad 0 \leq t < \infty \quad (12)$$

The mean and variance is defined as:

$$\mu_T = \frac{1}{\lambda}, \quad \sigma_T^2 = \frac{1}{\lambda^2} \quad (13)$$

2 Exercise 4.1.2

The diameter of a particle of contamination (in micrometers) is modeled with the probability density function $f(x) = 2/x^3$ for $x > 1$. Determine the following:

a. $P(X < 2)$

The probability is calculated using eq. 2 with the interval $[1; 2]$. This is the case even though the inequality is strictly smaller than 2. Since the probability is defined from an integral over some length, then $P(X = a) = \int_a^a f_X(x)dx = 0 \Rightarrow P(X < a) = P(X \leq a)$, $a \in \mathbb{R}$. Therefore the probability is:

$$P(X < 2) = \int_1^2 \frac{2}{x^3} dx = \int_1^2 2x^{-3} dx = 2 \frac{x^{-2}}{-2} \Big|_1^2 = -x^{-2} \Big|_1^2 = -\left(\frac{1}{2}\right)^2 + 1^2 = 0.75$$

b. $P(X > 5)$

The probability is calculated in the same way but now in the interval $[5; \infty[$ using WolframAlpha:

$$P(X > 5) = \int_5^\infty \frac{2}{x^3} dx = 0.04$$

c. $P(4 < X < 8)$

The probability is calculated in the interval using WolframAlpha:

$$P(4 < X < 8) = \int_4^8 \frac{2}{x^3} dx = 0.047$$

d. $P(X < 4 \text{ or } X > 8)$

Here using the superposition property of integrals the probability is:

$$P(X < 4 \text{ or } X > 8) = \int_1^4 \frac{2}{x^3} dx + \int_8^\infty \frac{2}{x^3} dx = 0.95$$

e. x such that $P(X < x) = 0.95$

This question asks us to find a value such that the probability of the value of the variable X being inside the interval $[1; x]$ is 95%. This can be written with eq. 2 as

follows and solved with WolframAlpha:

$$P(X < x) = \int_1^x \frac{2}{(x')^3} dx' = 0.95 \Rightarrow x = 2\sqrt{5}$$

3 Exercise 4.2.4

Determine the cumulative distribution function for the distribution in Exercise 4.1.2.

In order to find the cdf from the pdf given in exercise 4.1.2 we integrate in the defined domain:

$$F_X(x) = \int_1^x \frac{2}{(x')^3} dx' = -\frac{1}{(x')^2} \Big|_1^x = 1 - \frac{1}{x^2}$$

Which we can write up more general as:

$$F_X(x) = \begin{cases} 0, & x \leq 1 \\ 1 - \frac{1}{x^2}, & x > 1 \end{cases}$$

4 Exercise 4.3.1

Given the following probability density function, determine the mean and variance of X

$$f(x) = 1.5x^2, \quad \text{for } -1 < x < 1$$

From eq. 4 it can be seen that we will integrate an odd function $\mu_X \propto \int x^3$ over a symmetric interval $[-1; 1]$, which means that the positive and negative parts will cancel each other out. The result is then $\mu_X = 0$. This can also be proved by solving the integral:

$$\begin{aligned} \mu_X &= \int_a^b x f_X(x) dx = \int_{-1}^1 x \cdot 1.5x^2 dx = \int_{-1}^1 1.5x^3 dx = \frac{1.5}{4} x^4 \Big|_{-1}^1 = \frac{1.5}{4} [1^4 - (-1)^4] \\ &= \frac{1.5}{4} (1 - 1) = 0 \end{aligned}$$

For those doing quantum mechanics this is a very useful property of integrals to remember. It can save you a lot of time calculating complex integrals. Now the variance is calculated:

$$\sigma_X^2 = \int_{-1}^1 x^2 \cdot 1.5x^2 dx - \mu_X^2 = \frac{1.5}{5} x^5 \Big|_{-1}^1 - 0^2 = \frac{1.5}{5} [1^5 - (-1)^5] = \frac{3}{5} = 0.6$$

5 Exercise 4.4.1

Suppose X has a continuous uniform distribution over the interval $[-1, 1]$. Determine the following:

a. Mean, variance, and standard deviation of X .

As in exercise 4.3.1 because X is uniformly distributed over a symmetric interval the mean is $\mu_X = 0$. This can also be proved using eq. 8:

$$\mu_X = \frac{a+b}{2} = \frac{-1+1}{2} = 0$$

The variance of the uniformly distributed variable is:

$$\sigma_X^2 = \frac{(b-a)^2}{12} = \frac{[(-1)-1]^2}{12} = \frac{1}{3}$$

The standard deviation is calculated as:

$$\sigma_X = \sqrt{\sigma_X^2} = \sqrt{\frac{1}{3}}$$

b. Value for x such that $P(-x < X < x) = 0.90$.

From eq. 7 the pdf of a uniformly distributed continuous random variable is:

$$f_X(x) = \frac{1}{b-a} = \frac{1}{1-(-1)} = \frac{1}{2}$$

We can now write the problem as follows and solve with WolframAlpha:

$$P(-x < X < x) = \int_{-x}^x f_X(x') dx' = \int_{-x}^x \frac{1}{2} dx' = 0.9 \Rightarrow x = 0.9$$

c. Cumulative distribution function

The cdf is found by integration using eq. 3:

$$F_X(x) = \int_{-1}^x f_X(x') dx' = \int_{-1}^x \frac{1}{2} dx' = \frac{1}{2} x' \Big|_{-1}^x = \frac{1}{2}(x+1)$$

And in general then the cdf is defined as:

$$F_X(x) = \begin{cases} 0, & x \leq -1 \\ \frac{1}{2}(x+1), & -1 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

6 Exercise 4.5.1

Use Appendix Table III to determine the following probabilities for the standard normal random variable **Z**:

a. $P(Z < 1.32)$

The probability is found by integration using eq. 10 as:

$$P(Z < 1.32) = \int_{-\infty}^{1.32} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz = 0.91$$

b. $P(Z < 3)$

$$P(Z < 3) = \int_{-\infty}^3 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz = 0.999$$

c. $P(Z > 1.45)$

$$P(Z > 1.45) = \int_{1.45}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz = 0.074$$

d. $P(Z > -2.15)$

$$P(Z > -2.15) = \int_{-2.15}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz = 0.98$$

e. $P(-2.34 < Z < 1.76)$

$$P(-2.34 < Z < 1.76) = \int_{-2.34}^{1.76} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz = 0.95$$

7 Exercise 4.5.4

Assume that X is normally distributed with a mean of 10 and a standard deviation of 2. Determine the value for x that solves each of the following:

a. $P(X > x) = 0.5$

We use the standardization from eq. 11 such that:

$$P(X > x) = P\left(Z > \frac{X - \mu}{\sigma}\right) = P\left(Z > \frac{x - 10}{2}\right) = 1 - P\left(Z < \frac{x - 10}{2}\right) = 0.5$$

By subtracting 1 on each side and multiplying by -1 we get:

$$P\left(Z < \frac{x - 10}{2}\right) = 0.5$$

We use MATLAB to solve the normalized problem with "norminv(0.5)" which returns $P(Z < 0) = 0.5$ i.e. $z = 0$. We can now solve for x :

$$z = 0 = \frac{x - 10}{2} \Rightarrow x = 10$$

b. $P(X > x) = 0.95$

Using the same strategy as before:

$$P(X > x) = 1 - P\left(Z < \frac{x - 10}{2}\right) = 0.95 \Rightarrow P\left(Z < \frac{x - 10}{2}\right) = 0.05$$

Using MATLAB we find $z = -1.6449$ with "norminv(0.05)" meaning:

$$\frac{x - 10}{2} = -1.6449 \Rightarrow x = 6.71$$

c. $P(x < X < 10) = 0$

Again using the same strategy even though there is two bounds:

$$P(x < X < 10) = P\left(\frac{x - 10}{2} < X < \frac{10 - 10}{2}\right) = P\left(\frac{x - 10}{2} < X < 0\right) = 0$$

This can be written as:

$$P(x < X < 10) = P(X < 0) - P\left(X < \frac{x - 10}{2}\right) = 0.5 - P\left(X < \frac{x - 10}{2}\right) = 0$$

Rewriting gives:

$$P\left(X < \frac{x-10}{2}\right) = 0.5$$

This is solved with MATLAB as "norminv(0.5)" returning $z = 0$:

$$z = 0 = \frac{x-10}{2} \Rightarrow x = 10$$

d. $P(-x < X - 10 < x) = 0.95$

We can rewrite using the standardization as:

$$P(-x < X - 10 < x) = P(10 - x < X < x + 10) = P\left(\frac{(10 - x) - \mu}{\sigma} < Z < \frac{(10 + x) - \mu}{\sigma}\right) = 0.95$$

Inserting values:

$$P\left(\frac{(10 - x) - 10}{2} < Z < \frac{(10 + x) - 10}{2}\right) = P\left(-\frac{x}{2} < Z < \frac{x}{2}\right) = P\left(Z < \frac{x}{2}\right) - P\left(Z < -\frac{x}{2}\right) = 0.95$$

Since the normal distribution is symmetric we can use "norminv(0.025)" to find $0.025 = P(Z < -1.96)$ and "norminv(0.975)" to find $0.975 = P(Z < 1.96)$. We can now solve for x :

$$1.96 = \frac{x}{2} \Rightarrow x = 3.92$$

e. $P(-x < X - 10 < x) = 0.99$

Same idea as in d.:

$$P(-x < X - 10 < x) = P(10 - x < X < x + 10) = P\left(\frac{(10 - x) - \mu}{\sigma} < Z < \frac{(10 + x) - \mu}{\sigma}\right) = 0.99$$

Inserting:

$$P\left(\frac{(10 - x) - 10}{2} < Z < \frac{(10 + x) - 10}{2}\right) = P\left(-\frac{x}{2} < Z < \frac{x}{2}\right) = P\left(Z < \frac{x}{2}\right) - P\left(Z < -\frac{x}{2}\right) = 0.99$$

Solving with "norminv(0.005)" to find $0.005 = P(Z < -2.5758)$ and "norminv(0.995)" to find $0.995 = P(Z < 2.5758)$. We can now solve for x :

$$2.5758 = \frac{x}{2} \Rightarrow x = 5.1516$$

8 Exercise 4.5.13

The weight of a running shoe is normally distributed with a mean of 12 ounces and a standard deviation of 0.5 ounce.

a. What is the probability that a shoe weighs more than 13 ounces?

We first standardize:

$$P(X > 13) = P\left(Z > \frac{13 - 12}{0.5}\right) = P(Z > 2) = 1 - P(Z < 2)$$

This is solved in MATLAB using the following command:

$$P(X > 13) = 1 - \text{normcdf}(2) = 1 - 0.9772 = 0.028$$

Meaning there is a 2.28% probability for a running shoe to weigh more than 13 ounces.

b. What must the standard deviation of weight be in order for the company to state that 99.9% of its shoes weighs less than 13 ounces?

We use standardization:

$$P(X < 13) = 0.999 \Rightarrow P\left(Z < \frac{13 - 12}{\sigma}\right) = 0.999$$

Using "norminv(0.999)" gives $0.999 = P(Z < 3.09)$. We can now find the standard deviation needed for the claim:

$$3.09 = \frac{13 - 12}{\sigma} \Rightarrow \sigma = \frac{1}{3.09} = 0.324$$

c. If the standard deviation remains at 0.5 ounce, what must the mean weight be for the company to state that 99.9% of its shoes weighs less than 13 ounces?

We use the standardization since we know from before that $z = 3.09$:

$$Z = \frac{X - \mu}{\sigma} \Rightarrow 3.09 = \frac{13 - \mu}{0.5} \Rightarrow \mu = 11.46$$

9 Exercise 4.6.1

Suppose that X is a binomial random variable with $n = 200$ and $p = 0.4$. Approximate the following probabilities:

a. $P(X \leq 70)$

We know that the mean and variance of a binomial random variable can be found as:

$$\mu = np = 200 \cdot 0.4 = 80, \quad \sigma = \sqrt{np(1-p)} = \sqrt{200 \cdot 0.4 \cdot (1-0.4)} = 6.93$$

The probability is now found using standardization and calculated in MATLAB. We apply continuity correction to approximate the binomial distribution with a normal distribution (see page 80 in the book):

$$P(X \leq 70) \approx P\left(Z \leq \frac{70.5 - 80}{6.93}\right) = P(Z \leq -1.37) = \text{normcdf}(-1.37) = 0.0853$$

b. $P(70 < X < 90)$

Using the same strategy as before and solving with MATLAB:

$$\begin{aligned} P(70 < X < 90) &= P\left(\frac{70.5 - 80}{6.93} < Z < \frac{89.5 - 80}{6.93}\right) = P(-1.37 < Z < 1.37) \\ &= P(Z < 1.37) - P(Z < -1.37) = 0.91466 - 0.08534 = 0.8293 \end{aligned}$$

c. $P(X = 80)$

$$P(79.5 < X \leq 80.5) = P\left(\frac{79.5 - 80}{6.93} < Z < \frac{80.5 - 80}{6.93}\right) = P(-0.07217 < Z < 0.07217) = 0.0575$$

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10 Exercise 4.7.13

If the random variable has an exponential distribution with mean θ , determine the following:

a. $P(X > \theta)$

The probability is calculated using eq. 12 and the knowledge that the mean is defined as $\theta = 1/\lambda \Rightarrow \lambda = 1/\theta$:

$$P(X > \theta) = \int_{\theta}^{\infty} \frac{1}{\theta} e^{-\frac{1}{\theta}t} = \frac{1}{\theta} \frac{e^{-\frac{1}{\theta}t}}{-\frac{1}{\theta}} \Bigg|_{\theta}^{\infty} = -e^{-\frac{1}{\theta} \cdot \infty} + e^{-\frac{1}{\theta} \theta} = e^{-1} = 0.368$$

b. $P(X > 2\theta)$

Using the same strategy as before:

$$P(X > \theta) = \int_{2\theta}^{\infty} \frac{1}{\theta} e^{-\frac{1}{\theta}t} = \frac{1}{\theta} \frac{e^{-\frac{1}{\theta}t}}{-\frac{1}{\theta}} \bigg|_{2\theta}^{\infty} = -e^{-\frac{1}{\theta} \cdot \infty} + e^{-2\frac{1}{\theta}\theta} = e^{-2} = 0.135$$

c. $P(X > 3\theta)$

Using the same strategy as before:

$$P(X > \theta) = \int_{3\theta}^{\infty} \frac{1}{\theta} e^{-\frac{1}{\theta}t} = \frac{1}{\theta} \frac{e^{-\frac{1}{\theta}t}}{-\frac{1}{\theta}} \bigg|_{3\theta}^{\infty} = -e^{-\frac{1}{\theta} \cdot \infty} + e^{-3\frac{1}{\theta}\theta} = e^{-3} = 0.0498$$

d. How do the results depend on θ ?

We see that the result does not depend on θ .